

Task Manager REST API

Day 1 — Project Setup & Environment

1. Project Goal

The project is a resume-ready Task Manager REST API being developed step by step using Java and Spring Boot. The planned application will eventually support user registration and login with JWT, task CRUD, validation, user-specific task access, pagination, filtering, exception handling, and unit testing.

2. Development Environment Verified

Tool	Status / Version
Operating System	Windows 11 — Ready
Java	21.0.10 — Ready
javac	21.0.10 — Ready
Maven	3.9.16 — Ready
Git	2.53.0 — Ready
VS Code	Chosen development environment

3. Spring Boot Project Created

The project was created with Spring Initializr and opened in VS Code. The project uses Maven and Java 21.

Project location: F:\Projects\Task_Manager

Initial dependencies: Spring Web, Spring Data JPA, MySQL Driver, Validation, and Spring Security.

4. Important Initial Project Files

The initial project contains the Maven configuration (pom.xml), the Spring Boot main application class, application.properties, .gitignore, Maven wrapper files, and the src/test structure.

The layered packages such as controller, service, repository, entity, dto, security, and exception were intentionally not created yet. They will be introduced one layer at a time during later days.

5. Maven — What Was Learned

Maven is the Java project and dependency/build manager. The pom.xml file describes project information and the libraries required by the application. Maven downloads and manages those dependencies.

A Maven project must be run from the project root, where pom.xml exists.

An initial attempt was made from the Java package directory and produced a Maven plugin-prefix error. The project was then run from F:\Projects\Task_Manager, where pom.xml was found, and Maven proceeded correctly.

6. MySQL Database Setup

MySQL Community Server 8.0.43 was verified and a successful MySQL root login was performed.

Database created:

```
CREATE DATABASE taskmanager;
```

The database was selected with:

```
USE taskmanager;
```

SHOW TABLES returned an empty set. This is expected because no Java entities have been created yet.

7. application.properties — MySQL Connection

Spring Boot initially failed because JPA detected a database requirement but no DataSource URL/driver configuration was available. The application.properties file was then configured for MySQL.

The configuration used the taskmanager database on localhost:3306, the root MySQL user, the MySQL JDBC driver, Hibernate ddl-auto=update, SQL logging, and formatted SQL output.

Security lesson: the actual MySQL password should not be committed to GitHub. A safer environment-variable based configuration will be introduced before the repository is published.

8. First Successful Spring Boot Startup

After configuring MySQL, the application successfully connected to the database. The startup log confirmed MySQL 8.0.43, Hibernate, HikariCP, and Tomcat on port 8080. The application reported "Started TaskManagerApplication".

This means the REST server is running and waiting for HTTP requests. No custom API endpoint exists yet, so there is no application page or task data to display.

9. Spring Security Observation

Because Spring Security was added but custom authentication has not yet been configured, Spring Boot generated a default development login page and a temporary generated password. Random credentials were rejected as invalid.

This is not the final authentication system. Later, the project will replace the default behavior with user registration, password hashing, login, JWT generation, and JWT-protected task endpoints.

10. Basic Git Concepts Introduced

Git is the local version-control system; GitHub is the online hosting platform for repositories. The planned workflow is:

```
Working Directory → git add → Staging Area → git commit → Local Repository → git push → GitHub
```

Git/GitHub setup and the first GitHub push have not yet been completed. They are planned for the next step.

11. Codex Learning Plan

Codex will be introduced gradually rather than being asked to build the whole project automatically. The intended workflow is: inspect → understand → plan → modify → test → debug.

The first Codex activity will be a read-only inspection of the current project structure and configuration. No code changes should be requested until the project is understood.

12. Current Project Status at the End of Day 1

Area	Status
Java / Maven / VS Code	Completed
Spring Boot project	Completed
MySQL server verification	Completed
taskmanager database	Created
Database connection	Working
Spring Boot server	Successfully starts on port 8080
Custom REST endpoints	Not started
User entity	Not started
Task entity	Not started
JWT authentication	Not started
GitHub remote / push	Not started
Codex project workflow	Not started; planned next
Tests	Not started

Day 1 principle: Build slowly, understand each component, test each step, and use Git to preserve working versions.